

Ex. No.: 8

Experiment Title: Producer Consumer Using Semaphores

Aim:

To write a program to implement a solution to the Producer-Consumer problem using semaphores.

Algorithm:

1. Initialize semaphores: `empty`, `full`, and `mutex`.
2. Create two threads – a producer thread and a consumer thread.
3. Wait for each thread's termination using `pthread_join()`.
4. In producer: call `sem_wait()` on `empty` and `mutex` before entering the critical section.
5. Produce an item in the critical section.
6. Call `sem_post()` on `mutex` and then on `full` after exiting.
7. In consumer: call `sem_wait()` on `full` and `mutex` before entering the critical section.
8. Consume an item in the critical section.
9. Call `sem_post()` on `mutex` and then on `empty`.
10. Loop the producer and consumer threads 10 times each, then terminate.

Program Code

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
```

```

#include <unistd.h>

#define SIZE 5

int buffer[SIZE];
int in = 0, out = 0, item = 0;

sem_t empty, full, mutex;

void *producer(void *arg) {
    for(int i = 0; i < 10; i++) {
        sem_wait(&empty);
        sem_wait(&mutex);

        item++;
        buffer[in] = item;
        printf("Producer produces item %d at buffer[%d]\n", item, in);
        in = (in + 1) % SIZE;

        sem_post(&mutex);
        sem_post(&full);
        sleep(1);
    }
    pthread_exit(NULL);
}

void *consumer(void *arg) {
    for(int i = 0; i < 10; i++) {
        sem_wait(&full);
        sem_wait(&mutex);

        int consumed = buffer[out];
        printf("Consumer consumes item %d from buffer[%d]\n",
consumed, out);
        out = (out + 1) % SIZE;

        sem_post(&mutex);
        sem_post(&empty);
    }
}

```

```

        sleep(1);
    }
    pthread_exit(NULL);
}

int main() {
    pthread_t prod, cons;

    sem_init(&empty, 0, SIZE);
    sem_init(&full, 0, 0);
    sem_init(&mutex, 0, 1);

    pthread_create(&prod, NULL, producer, NULL);
    pthread_create(&cons, NULL, consumer, NULL);

    pthread_join(prod, NULL);
    pthread_join(cons, NULL);

    sem_destroy(&empty);
    sem_destroy(&full);
    sem_destroy(&mutex);

    return 0;
}

```

Sample Output:

```

Producer produces item 1 at buffer[0]
Consumer consumes item 1 from buffer[0]
Consumer waits (buffer empty)...
Producer produces item 2 at buffer[1]
Producer produces item 3 at buffer[2]
Producer produces item 4 at buffer[3]
Producer produces item 5 at buffer[4]
Producer waits (buffer full)...
Consumer consumes item 2 from buffer[1]
Consumer consumes item 3 from buffer[2]

```

```
Consumer consumes item 4 from buffer[3]  
Consumer consumes item 5 from buffer[4]
```

Result:

The program successfully implements the Producer-Consumer problem using semaphores with correct synchronization between producer and consumer threads.